\'ETALE-SMOOTH LOCAL ON THE SOURCE-AND-TARGET FROM THREE LOCAL PROPERTIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The pinned source reads 'with X_i a scheme' at proof line 3367 where X' is meant; delivered verbatim per policy, flagged here for the reviewer.

Allowed standard prerequisites (editorial).

- Definition of \'etale-smooth local on the source-and-target. Stacks Project, Descent on Spaces, Definition definition-local-source-target. Supplies the target notion, in particular part (3)(b) whose data the proof assembles.
- Stability under \'etale precomposition. Stacks Project, Descent on Spaces, Lemma lemma-precompose-property-local-source. Derives stability under precomposing with \'etale morphisms from assumption (1).
- Stability under smooth base change. Stacks Project, Descent on Spaces, Lemma lemma-pullback-property-local-target. Derives stability under smooth base change from assumption (2).
- Scheme-side \'etale-smooth criterion. Stacks Project, Descent, Lemma descent-lemma-etale-tau-local-source-target. Concludes for morphisms of schemes after the reduction.

Source reference. Original lemma 06EQ: sources/stacks-chow/tags.txt, source lines 8224–8224. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 06EM: sources/stacks-chow/tags.txt, source lines 8221–8221. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 06F8: sources/stacks-chow/tags.txt, source lines 8242–8242. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0CF1: sources/stacks-chow/tags.txt, source lines 15596–15596. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces over S. Assume*

(1) *$\mathcal{P}$ is étale local on the source,*
(2) *$\mathcal{P}$ is smooth local on the target, and*
(3) *$\mathcal{P}$ is stable under postcomposing with open immersions: if $f : X \to Y$ has $\mathcal{P}$ and $Y \subset Z$ is an open embedding then $X \to Z$ has $\mathcal{P}$.*

Then $\mathcal{P}$ is étale-smooth local on the source-and-target.

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Proof. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces which satisfies conditions (1), (2) and (3) of the lemma. By Lemma 06EQ we see that $\mathcal{P}$ is stable under precomposing with étale morphisms. By Lemma 06EM we see that $\mathcal{P}$ is stable under smooth base change. Hence it suffices to prove part (3) of Definition 06F8 holds.

More precisely, suppose that $f : X \to Y$ is a morphism of algebraic spaces over S which satisfies Definition 06F8 part (3)(b). In other words, for every $x \in X$ there exists a smooth morphism $b_x : V_x \to Y$, an étale morphism $U_x \to V_x \times_Y X$, and a point $u_x \in |U_x|$ mapping to x such that $h_x : U_x \to V_x$ has $\mathcal{P}$. The proof of the lemma is complete once we show that f has $\mathcal{P}$.

Let $a_x : U_x \to X$ be the composition $U_x \to V_x \times_Y X \to X$. Set $U = \coprod U_x$, $a = \coprod a_x$, $V = \coprod V_x$, $b = \coprod b_x$, and $h = \coprod h_x$. We obtain a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow[h]{} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

with b smooth, $U \to V \times_Y X$ étale, a surjective. Note that h has $\mathcal{P}$ as each h_x does and $\mathcal{P}$ is smooth local on the target. In the next paragraph we prove that we may assume U, V, X, Y are schemes; we encourage the reader to skip it.

Let X, Y, U, V, a, b, f, h be as in the previous paragraph. We have to show f has $\mathcal{P}$. Let $X' \to X$ be a surjective étale morphism with X_i a scheme. Set $U' = X' \times_X U$. Then $U' \to X'$ is surjective and $U' \to X' \times_Y V$ is étale. Since $\mathcal{P}$ is étale local on the source, we see that $U' \to V$ has $\mathcal{P}$ and that it suffices to show that $X' \to Y$ has $\mathcal{P}$. In other words, we may assume that X is a scheme. Next, choose a surjective étale morphism $Y' \to Y$ with Y' a scheme. Set $V' = V \times_Y Y'$, $X' = X \times_Y Y'$, and $U' = U \times_Y Y'$. Then $U' \to X'$ is surjective and $U' \to X' \times_{Y'} V'$ is étale. Since $\mathcal{P}$ is smooth local on the target, we see that $U' \to V'$ has $\mathcal{P}$ and that it suffices to prove $X' \to Y'$ has $\mathcal{P}$. Thus we may assume both X and Y are schemes. Choose a surjective étale morphism $V' \to V$ with V' a scheme. Set $U' = U \times_V V'$. Then $U' \to X$ is surjective and $U' \to X \times_Y V'$ is étale. Since $\mathcal{P}$ is smooth local on the source, we see that $U' \to V'$ has $\mathcal{P}$. Thus we may replace U, V by U', V' and assume X, Y, V are schemes. Finally, we replace U by a scheme surjective étale over U and we see that we may assume U, V, X, Y are all schemes.

If U, V, X, Y are schemes, then f has $\mathcal{P}$ by Descent, Lemma 0CF1. □